

56 Chapter Review



➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

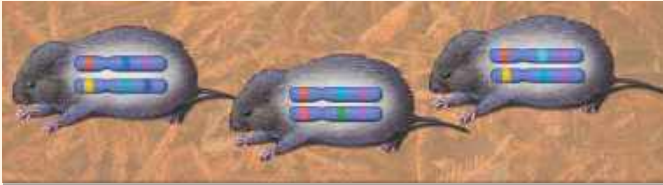
SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkz9t.

CONCEPT 56.1

Human activities threaten earth's biodiversity (pp. 1261–1266)

- Biodiversity can be considered at three main levels:



Genetic diversity: enables adaptation to environmental change



Species diversity: maintains communities and food webs



Ecosystem diversity: provides life-sustaining services

- Our biophilia enables us to recognize the value of biodiversity for its own sake. Other species also provide humans with food, fiber, medicines, and **ecosystem services**.
- Four major threats to biodiversity are habitat loss, **introduced species**, overharvesting, and global change.

? Give at least three examples of key ecosystem services that nature provides for people.

CONCEPT 56.2

Population conservation focuses on population size, genetic diversity, and critical habitat (pp. 1266–1270)

- When a population drops below a **minimum viable population (MVP)** size, its loss of genetic variation due to inbreeding and genetic drift can trap it in an **extinction vortex**.
- A loss of critical habitat can cause threatened populations to show a downward trend, even if the population is far above its minimum viable population size.
- Conserving species often requires resolving conflicts between the habitat needs of **endangered species** and human demands.

? Why is the minimum viable population size smaller for a genetically diverse population than for a less genetically diverse population?

CONCEPT 56.3

Landscape and regional conservation help sustain biodiversity (pp. 1270–1274)

- The structure of a landscape can strongly influence biodiversity. As habitat fragmentation increases and edges become more extensive, biodiversity tends to decrease. **Movement corridors** can promote dispersal and help sustain populations.
- **Biodiversity hot spots** are also hot spots of extinction and thus prime candidates for protection. Sustaining biodiversity in parks and reserves requires management to ensure that human activities in the surrounding landscape do not harm the protected habitats. The **zoned reserve** model recognizes that conservation efforts often involve working in landscapes that are greatly affected by human activity.
- **Urban ecology** is the study of organisms and their environment in primarily urban settings.

? Give two examples that show how habitat fragmentation can harm species in the long term.

CONCEPT 56.4

Earth is changing rapidly as a result of human actions (pp. 1274–1284)

- Agriculture removes plant nutrients from ecosystems, so large supplements are usually required. The nutrients in fertilizer can pollute groundwater and surface-water aquatic ecosystems (eutrophication), where they can stimulate excess algal growth.
- The release of industrial compounds, pesticides, pharmaceuticals, and plastic waste has polluted the environment with harmful substances that often persist for long periods and become increasingly concentrated in successively higher trophic levels of food webs (**biological magnification**).
- Because of the burning of fossil fuels and other human activities, the atmospheric concentration of CO₂ and other greenhouse gases has been steadily increasing. These increases have caused **climate change**, including significant global warming and changing patterns of precipitation. Climate change has already affected many ecosystems.
- The ozone layer reduces the penetration of UV radiation through the atmosphere. Human activities, notably the release of chlorine-containing pollutants, have eroded the ozone layer, but government policies are helping to solve the problem.

? Thinking about biological magnification of toxins, is it healthier to feed at a lower or higher trophic level? Explain.

CONCEPT 56.5

Sustainable development can improve human lives while conserving biodiversity (pp. 1284–1285)

- **Sustainable development** refers to economic development that meets the needs of people today without limiting the ability of future generations to meet their needs.
- Costa Rica's success in conserving tropical biodiversity has involved a partnership among the government, other organizations, and private citizens. Human living conditions in Costa Rica have improved along with ecological conservation.

? Why is sustainability such an important goal for conservation biologists?